

Validity Analysis: DRBENCHMARK

Target context: EU Pharmaceutical Regulatory NLP — French

Overall risk: HIGH

Deployment Context

Use case: In a pharmaceutical setting, a regulatory specialist uses document management software to ensure compliance in French drug labeling and scientific abstracts by applying a model evaluated on semantic textual similarity and fine-grained named-entity recognition. The system identifies over a dozen types of precise entities, such as chemical compounds and dosage modes, while verifying that safety warnings across different leaflets or patents are semantically consistent. These results determine whether pharmacological documentation meets strict professional standards for official submission or requires manual revision to prevent regulatory non-compliance.

Target population: Regulatory affairs specialists, pharmacologists, and legal experts at pharmaceutical companies or government health agencies.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	2	Significant gaps	high
Input Content 	2	Significant gaps	high
Input Form 	4	Minor gaps	high
Output Ontology 	1	Serious concern	high
Output Content 	2	Significant gaps	high
Output Form 	4	Minor gaps	high
Average	2.5		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

DrBenchmark is the largest French biomedical NLU benchmark and provides genuine value for general French biomedical model selection [Q83], but it is severely misaligned with the EU pharmaceutical regulatory NLP deployment on the four HIGH-priority dimensions identified in the elicitation. Output Ontology (score 1) is the most critical concern: NER schemas use

clinical UMLS Semantic Groups that conflate INN, excipients, and brand names under CHEM (QUAERO-D15, MANTRAGSC-D15), provide no ATC code class [WEB-11], no contraindication qualifier class (QUAERO-D21, PXCORPUS-D15), and systematically O-tag dose values (QUAERO-D18, E3C-D7); STS is calibrated for general semantic proximity rather than regulatory equivalence, demonstrably treating legally material differences as near-synonymy (CLISTER-D13: 2× numeric difference scored 4.0; DEFT2020-D18: dropped 'congénitale' scored 4.6). Input Ontology (score 2) and Input Content (score 2) reflect the absence of SmPC, CTD module, and EU-RMP genres — QUAERO EMEA's ~10 segmented PIL documents [WEB-3, WEB-4] are the only confirmed regulatory-genre source. Output Content (score 2) reflects systematically misaligned annotator populations (academic NLP and clinical, no regulatory-affairs specialists [Q8]) with empirical variance evidence (DEFT2020-D10, DEFT2020-D22). Input Form (score 4) and Output Form (score 4) are well-aligned — both deployment and benchmark are text-only French Latin-script with compatible output representations. The benchmark can support model selection at the foundational French-biomedical-LM level, but cannot validate deployment-grade regulatory NER or regulatory-equivalence STS without substantial supplementation.

Practical Guidance

What This Benchmark Measures

DrBenchmark measures general French biomedical language understanding across NER, STS, classification, MCQA, and POS tagging on clinical, research, and pharmacy-exam text. For this deployment, it provides useful signal on (a) which French biomedical MLM has the best general performance profile (DrBERT-FS, DrBERT-CP, CamemBERT-bio per [Q60–Q62]), (b) coarse-grained pharmaceutical NER on EMEA-genre patient leaflets via QUAERO (QUAERO-D1–D15), and (c) baseline French biomedical STS variance behavior via CLISTER and DEFT2020. The strongest dimensions for this deployment are Input Form and Output Form — both at score 4.

Construct Depth

The benchmark probes general French biomedical NLU constructs at moderate depth, but does not probe the regulatory-specific constructs the deployment requires. NER probes clinical entity recognition (UMLS Semantic Groups, MeSH chapters, ICD-10 chapters) — not regulatory entity recognition (INN as distinct from CHEM, ATC codes, excipient names, EMA-templated posology, contraindication qualifiers). STS probes general clinical sentence similarity — not regulatory equivalence. Empirically, pharmaceutical content is broadly present in the inputs but is systematically unannotated or under-annotated (E3C-D7, QUAERO-D18, CAS-D13). The benchmark's 75% top-task win rate for domain-specialized models [Q62] does not translate to regulatory-grade precision (per [WEB-12]: general biomedical NLP systems require modification for regulatory use).

What Else You Need

To complete this assessment for regulatory deployment, the user needs: (1) An OO supplement — a regulatory entity schema (INN, ATC, excipient, posology template, contraindication qualifier) and a regulatory-equivalence STS rubric calibrated for dose-threshold and population-qualifier sensitivity; neither exists in published literature (gap_id 3 confirmed). (2) An IO/IC supplement — a held-out evaluation set of authentic SmPC, CTD module narrative, and EU-RMP text under QRD v10.4 [WEB-7], since QUAERO EMEA covers only PIL register and ~10 segmented documents. (3) An OC supplement — re-annotation of a sample of QUAERO EMEA and CLISTER pairs by regulatory-affairs specialists to quantify the gap between biomedical NLP annotation norms and EMA/ANSM judgments (anticipated per A4). (4) Empirical evaluation against the Kadi et al. 2025 SmPC IDMP extraction baseline [WEB-11] for ATC and excipient extraction, which is field-wide unsolved.